

Induction

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Exercise 1

$$P(n) : S_n = \frac{n(n+1)(2n+1)}{6}$$

Claim: $P(n)$ is true $\forall n \in \mathbb{Z}^+$

Proof: (By induction)

Base Case: $P(1)$ is true because:

$$1^2 = \frac{1(1+1)(2 \cdot 1 + 1)}{6} = \frac{1 \cdot 2 \cdot 3}{6} = 1$$

.

Induction Hypothesis: Assume $P(k)$ is true for some $k \in \mathbb{Z}^+$. i.e. $s_k = 1^2 + 2^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$.

We know that:

$$1^2 + 2^2 + \dots + k^2 + (k+1)^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \quad (1)$$

$$= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} \quad (2)$$

$$= \frac{(k+1)[k(2k+1) + 6(k+1)]}{6} \quad (3)$$

$$= \frac{(k+1)[2k^2 + 7k + 6]}{6} \quad (4)$$

$$= \frac{(k+1)(k+2)(2k+3)}{6} \quad (5)$$

Thus $P(k+1)$ is true. By PMI, $P(n)$ is true $\forall n \in \mathbb{Z}^+$. ■

Exercise 2

$$t_n = 1 + 2 + \dots + n = t_{n-1} + n$$

These are just two separate ways to write out t_n . Below is what will be part of our statement:

$$P(n) : t_n = \frac{n(n+1)}{2}$$

What are we doing?

$$n = 1 : \frac{1(1+1)}{2} = 1 \tag{6}$$

$$n = 2 : \frac{2(2+1)}{2} = 3 \tag{7}$$

$$n = 3 : \frac{3(3+1)}{2} = 6 \tag{8}$$

$$\tag{9}$$

We want to show that this equals t_n .

Claim: $P(n)$ is true $\forall n \in \mathbb{Z}^+$.

Proof: (By Induction)

$P(1)$ is true, since $1 = \frac{1(1+1)}{2} = 1$.

Induction Hypothesis: Assume $P(k)$ is true for **some** $k \in \mathbb{Z}^+$.

Thus, $t_k = \frac{k(k+1)}{2}$.

Aside: We need to show that $t_{k+1} = \frac{(k+1)(k+2)}{2}$.

Return to proof:

We know

$$t_{k+1} = t_k + (k+1) \tag{10}$$

$$= \frac{k(k+1)}{2} + (k+1) \tag{11}$$

$$= \frac{k(k+1) + 2(k+1)}{2} \tag{12}$$

$$= \frac{(k+1)[k+2]}{2} \tag{13}$$

Thus, $P(k+1)$ is true. So by PMI, $P(n)$ is true $\forall n \in \mathbb{Z}^+$. ■